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Rigid body mechanics

Problem Set VII

Exercise 1:

A homogeneous solid sphere S, of mass m, radius a, and center G, rolls without slipping on a fixed horizontal plane Π, located at elevation $z = 0$, relative to a frame $R(O, X, Y, Z)$, with basis $(\vec{i}, \vec{j}, \vec{k})$, assumed to be Galilean. The OZ axis is vertical upward. We denote by H the intersection point of plane Π with the OZ axis. We denote by I the geometric contact point of S with Π and $\vec{\Omega}$ the instantaneous rotation vector of S with respect to R. The torsor of contact actions at point I is reduced to the reaction $\vec{R}$.

1. Write the rolling without slipping relation of S on plane Π, then deduce the velocity and acceleration vectors of the center of mass G, with respect to R, as a function of $\vec{\Omega}$ and $\dot{\vec{\Omega}}$.

2. Determine the time-dependent vector equation of motion of G, $\overrightarrow{HG}(t)$, by applying the general theorems. What is the nature of the motion of G? We denote by $\vec{v}_0$ the initial velocity vector of G and by $\vec{r}_0$ the initial position vector of G.

In what follows, we assume that plane Π is rotating about the OZ axis and we denote by $\vec{\omega}$ the instantaneous rotation vector of the plane with respect to R.

3. Determine the velocity of center of mass G, $\vec{v}_G$, using the rolling without slipping relation.

4. Find, using the general theorems, the acceleration of center of mass G, $\vec{\gamma}_G$, as a function of $\vec{\omega}$ and $\vec{\Omega}$ and their time derivatives. Deduce a first vector integral of motion.

Let r and φ be the polar coordinates of G, defined in the plane XOY by $\overrightarrow{HG} = r\vec{u}_r$. Initially, we have the following values $r_0 = r(0)$, $\dot{r}_0 = \dot{r}(0)$, and $\dot{\varphi}_0 = \dot{\varphi}(0)$. And we denote by μ the sliding friction coefficient of S on plane Π.

5. Determine the time-dependent equations of motion $r(t)$ and $\varphi(t)$, and deduce $\overrightarrow{HG}(t)$ as a function of time.

6. Determine $\vec{R}$ and $\vec{M}_I$ and indicate the condition that μ must satisfy for rolling without slipping to persist.

Correction